
Supply & Demand

OpenStax Principles of Economics 3e, Sections 3.1–3.3

1 Core Concepts

Economists use the term **demand** to refer to the amount of a good or service that consumers are willing and able to purchase at each price. Both conditions matter — willingness without the ability to pay does not count as **effective demand**.

The **law of demand** states that when the price of a good rises, the quantity demanded falls, and when price falls, quantity demanded rises. This inverse relationship produces a downward-sloping **demand curve**, with price on the vertical axis and quantity on the horizontal axis.

Supply is the amount of a good or service that producers are willing to sell at each price. The **law of supply** states that when price rises, quantity supplied rises, and when price falls, quantity supplied falls. This positive relationship produces an upward-sloping **supply curve** on the same axes.

One distinction is worth keeping in mind throughout: **demand** refers to the entire curve, while **quantity demanded** refers to a specific point on it. The same applies to supply and quantity supplied. A change in price causes movement along an existing curve; only a change in something other than price shifts the whole curve.

When the demand and supply curves are placed on the same diagram, they intersect at a single point called **equilibrium**. The **equilibrium price** is the only price at which the quantity consumers want to buy exactly equals the quantity producers want to sell; that amount is called the **equilibrium quantity**.

If the market price sits above equilibrium, quantity supplied exceeds quantity demanded, creating a **surplus**; sellers lower prices to move inventory, pushing the market back toward equilibrium. If price falls below equilibrium, quantity demanded exceeds quantity supplied, creating a **shortage**; buyers bid prices back up. Markets self-correct toward equilibrium through these pressures.

Key distinction: A *movement along* a curve happens when **price** changes. A *shift of the entire curve* happens when something **other than price** changes.

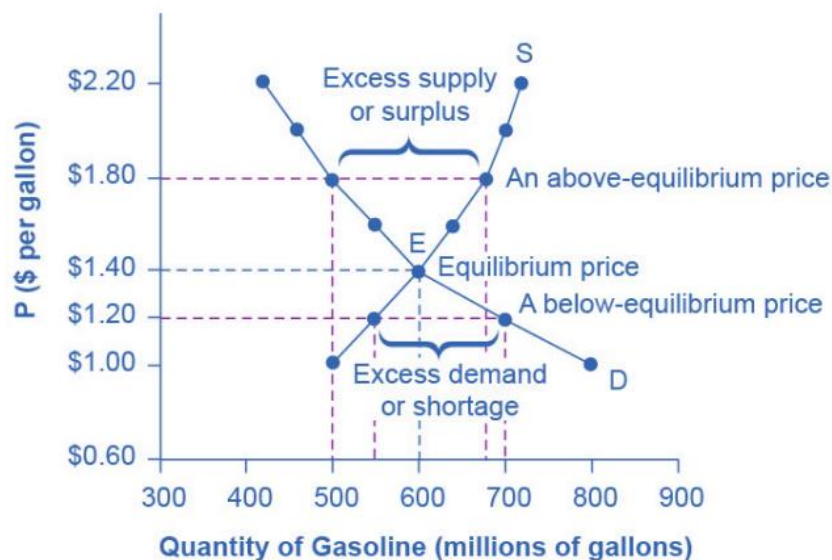


FIGURE 1 Gasoline Supply and Demand: The demand (D) and supply (S) curves intersect at equilibrium point E (\$1.40, 600), where quantity demanded equals, quantity supplied. Prices above equilibrium create excess supply, while prices below create excess demand.

2 What Shifts Demand?

Because a demand curve holds all other factors constant (economists call this assumption *ceteris paribus*, meaning "all else equal"), a change in any of those other factors will shift the entire curve to a new position. Six factors commonly shift demand. Income is one: for most goods, called normal goods, demand rises when incomes rise and falls when incomes fall. Tastes and preferences matter too — if consumers become more enthusiastic about a product, demand increases; if they lose interest, it decreases. Population size and composition affect demand because more potential buyers means more demand overall. The prices of related goods also play a role: a substitute is a good that can replace another, so if a substitute becomes more expensive, demand for the original good rises; a complement is a good used alongside another, so if a complement becomes cheaper, demand for the paired good rises. Finally, expectations about future prices can shift demand today — if consumers expect prices to rise soon, they may buy more now.

A rightward shift means more is demanded at every price; a leftward shift means less is demanded at every price.

Postal service example: Email and social media are substitutes for mail. As they become more attractive, consumer tastes shift away from postal services → demand curve shifts **left**.

3 What Shifts Supply?

Supply = the whole curve. *Quantity supplied* = one point on it. Same rule: only non-price factors shift the curve itself.

The same logic applies to supply. If something other than price changes the conditions under which producers operate, the entire supply curve shifts. Input costs are the most direct example: when the wages or materials a firm must pay increase, profits shrink and firms supply less at every price, shifting supply left. Conversely, falling input costs shift supply right. Improvements in technology lower the cost of production and therefore increase supply. Natural conditions matter for many industries — good weather increases agricultural supply, bad weather reduces it. Government policies such as taxes raise costs and reduce supply, while subsidies lower costs and increase supply. Finally, when more firms enter a market, the total quantity supplied rises; when firms exit, it falls.

*Postal service example: Rising postal worker wages are a production cost increase → supply curve shifts **left**.*

4 The Four-Step Process

Step 1. Draw the initial supply and demand graph and identify the original equilibrium price and quantity.

Step 2. Does the event affect **supply** or **demand**?

Step 3. Does that curve shift **right** or **left**?

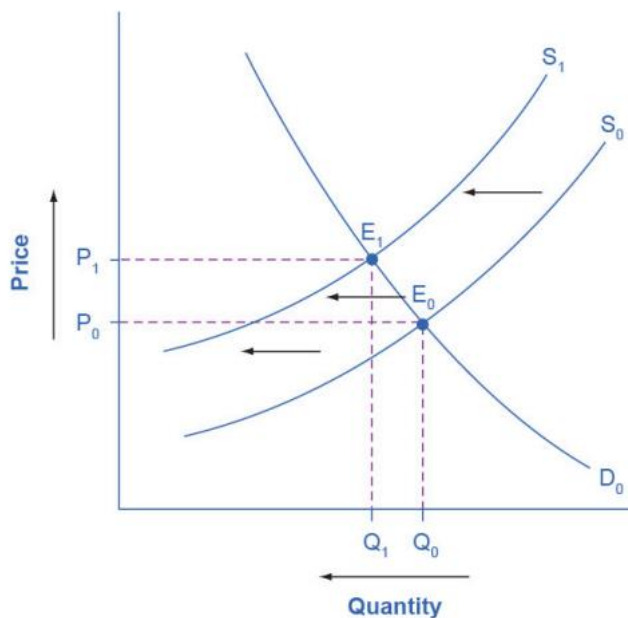
Step 4. Find the **new equilibrium** and compare the changes in price and quantity.

5 Worked Example — The U.S. Postal Service

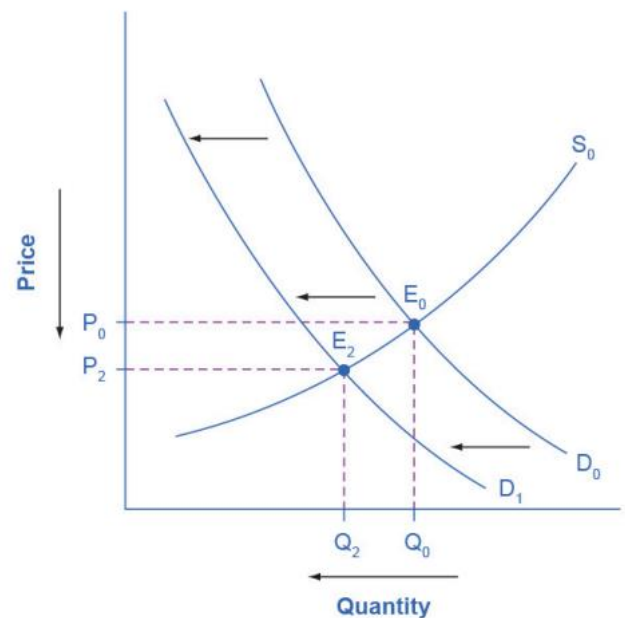
The postal service faces two simultaneous pressures. Rising worker wages are an input cost, which means higher production costs for the postal service at every level of output. Working through the four steps: this is a supply-side event, wages are a cost of production, higher costs reduce what the postal service is willing to supply at any given price, so the supply curve shifts left.

The result is a lower equilibrium quantity and a higher equilibrium price.

At the same time, more people are switching to email, texting, and social media platforms. These are substitutes for traditional mail. Working through the four steps again: this is a demand-side event, the shift toward digital communication reflects a change in tastes away from postal services, so the demand curve shifts left. The result is a lower equilibrium quantity and a lower equilibrium price.



(a) Shift in supply



(b) Shift in demand

FIGURE 1 Higher Compensation for Postal Workers: A Four-Step Analysis (a) Higher labor compensation causes a leftward shift in the supply curve, a decrease in the equilibrium quantity, and an increase in the equilibrium price. (b) A change in tastes away from Postal Services causes a leftward shift in the demand curve, a decrease in the equilibrium quantity, and a decrease in the equilibrium price.

Combining both analyses: because both curves shift left, the effect on quantity is unambiguous — the equilibrium quantity of postal services definitely decreases. The effect on price, however, is ambiguous. The leftward supply shift pushes price up; the leftward demand shift pushes price down. Without knowing which effect is larger, we cannot determine whether the final price is higher or lower than the original. This is a common outcome when both curves shift — one of the two outcomes (price or quantity) will be determinate, and the other will be ambiguous.

Quick Reference

Demand curve: downward sloping; price $\uparrow \rightarrow Q_d \downarrow$

Supply curve: upward sloping; price $\uparrow \rightarrow Q_s \uparrow$

Equilibrium: where curves cross; surplus above, shortage below

Demand shifters: income, tastes, population, substitutes, complements, expectations

Supply shifters: input costs, technology, natural conditions, government policy, number of producers

Price change $\rightarrow$ movement *along* a curve

Non-price change $\rightarrow$ *shift* of the curve

Both curves shift left $\rightarrow$ quantity definitely $\downarrow$, price direction ambiguous